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Systematic review

Systematic assessment of clinical outcomes and toxicities of proton radiotherapy for reirradiation

Vivek Verma [a], Jean-Claude M. Rwigema [b], Robert S. Malyapa [c], William F. Regine [c], Charles B. Simone II [c,*]

[a] Department of Radiation Oncology, University of Nebraska Medical Center, Omaha; [b] Department of Radiation Oncology, Mayo Clinic, Scottsdale; and [c] Department of Radiation Oncology, University of Maryland School of Medicine, Baltimore, USA

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ABSTRACT

Reirradiation (reRT) for locoregional recurrences poses unique challenges and risks; re-treatment using proton beam radiotherapy (PBT) could prove advantageous. Assessing clinical outcomes and toxicity profiles, this systematic review comprehensively evaluated available evidence regarding PBT reRT. Fourteen original investigations across central nervous system (CNS) ($n = 6$), head/neck (H&N) ($n = 4$), lung ($n = 2$), and gastrointestinal ($n = 2$) malignancies were analyzed. PBT for recurrent uveal melanoma achieved 5-year eye retention of 55%; for chordomas, reRT afforded a 2-year local control and overall survival (OS) of 85% and 80%, respectively. Multiple PBT reRT studies for adult gliomas illustrate no grade $\geq$3 toxicities. Two pediatric CNS tumor studies demonstrated the safety and efficacy of reRT, with one total grade 3 toxicity and achievement of longer-term OS. PBT for H&N malignancies shows appropriate local/locoregional control and favorable toxicity profiles versus historical photon-based methods, including low (9–10%) rates of feeding tube placement. PBT for recurrent lung cancer can achieve favorable survival with expected toxicities/complications of reRT, especially with concurrent chemotherapy and centrally located recurrences. Lastly, PBT reRT in gastrointestinal malignancies induced very few high-grade complications. Hence, based on the limited existing data, PBT is a notably safe reRT modality for effective salvage of recurrent disease. Institutional experiences must continue to be reported: dosimetric correlations, late toxicities, and advanced PBT techniques.

The potential of proton beam therapy (PBT) has transitioned from dosimetric advantages to encouraging results of phase I and II trials. Despite no completed phase III trials comparing PBT to photon-based techniques, the promise of PBT has been exemplified by a sharp rise in facilities installed across the globe. Compared with photon therapy, PBT can achieve decreased proximal dose deposition and virtually no exit dose distal to the target volume [1]. As a result of this relative sparing of radiation doses to nearby organs-at-risk (OARs), PBT may allow for clinical toxicity reductions. The decreased integral dose has also been postulated to induce fewer secondary neoplasms than with photons [2,3]. Increasing data demonstrate PBT to be a safe and efficacious option to treat many neoplasms [4–12], with PBT being relatively most advantageous for cases where normal tissue receives “clinically important” radiation doses [13].

Patients with local or regional recurrences that are in the field of a prior radiotherapy treatment course are currently a treatment dilemma. Local therapy with surgery or reirradiation (reRT) may provide the best chance of long-term disease control and the best chance at a potential cure, but surgery after prior radiation is often not possible and reRT with photon-based radiotherapy is associated with high rates of complications and morbidities that can negatively impact overall health status, quality of life, and even survival [14]. Given the potentially grave risks with photon reRT, patients with locoregional recurrences are generally treated with systemic therapy alone [15] and thus generally do not receive potentially curative therapy.

While re-treatment of locoregional recurrences may be a safer option when delivered with various advanced radiotherapy (RT) modalities, data on their safety and efficacy are currently lacking. The high conformality of PBT offers substantial potential in the reRT setting to decrease normal tissue doses for tissues that have received “clinically important” radiation doses during the first radiotherapy treatment course. PBT reRT, therefore, could result in clinically meaningful reductions in toxicities and complications, potentially allowing for patients to maintain quality of life and/or functional status while receiving potentially curative therapy. As such, reviewing clinical results and patient selection of new technologies in this setting is essential to justify their further use and

* Corresponding author at: University of Maryland School of Medicine, Department of Radiation Oncology, Maryland Proton Treatment Center, 850 W. Baltimore St., Baltimore, MD 21201, USA.

E-mail address: charlessimone@umm.edu (C.B. Simone II).

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